SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1703**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I K.Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Pujitha

18/20

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Uniform cost search (UCS)

Definition

Uniform Cost search Expands the Path with the lowest total Cost first.

Working:

- Starts from the intial position
- Calculates the total Path Cost
- Always select the least cost Path
- stops when Exit is Reached.

~~Dis~~Advantages:

* High memory usuage.
* Can be slow for large mazes.

Advantages:

* finds the optimal Path
* works well when Path Cost is different.

Complexity:

* Time : Exponential (depends on branching factor and cost)
* Space : Exponential

(2) Iterative Deepening search (IDS)

Definition:

IDS repeadtlely performs Depth-first search with increasing Depth limits.

Working:

- Search depth = 0
- Then depth = 1
- Then depth = 2
- Continous until the Exit is found.

Advantages:

* Requires very little memory.
* Complete for finite search spaces.
* Finds shallow solutions Efficiently.

Disadvantages:

* Repeats searches
* May take more time because nodes are revisited.

Complexity:

* Time : $O(b^d)$
* Space : $O(b \times d)$

where:- b = branching factor
d = depth of solution

Relation to Intelligent Agents:

1) Simple Reflex Agent

* Reacts only to Current state

* Cannot Plan a Path
* Not suitaible for maze solving.

2) Model- Based Agent :

* Remember visited locations
* Avoids Revisiting the same Path
* Better than simple Reflex.

3) Goal- Based Agent

* Goal is to Reach the maze Exit
* Uses search Algorithm like UCS or IDS
* Plans the Path before moving.

4) Utility - Based Agent

* Choses the Best Action
  → Time
  → Energy
  → Safety
* Useful when multiple Exits or Path Costs Exists.

Best combination :

Goal- Based Agent + Uniform cost search (UCS)

Justification :-

* Goal - Based Agent focuses on reaching the Exit

• UCS gaurantees the least Cost-Path.
• Suitable when movement Costs vary.

If memory limited, then :-

Goal Based-Agent + Iterative Deeping search (IDS)

→ It is better choice because IDS requires much less memory.

Conclusion:

* UCS is best for finding the optimal Path
* IDS is best when memory is limited.
* Goal Based-Agent Combined with UCS which. provides effective and intelligent maze navigation.

1) INTRODUCTION:

A smart home Controls the lightening temperature and sensors such as Motion detectors, temperator sensor and other input data. It learns users preferences to provide Comfort and security.

Relating to Intelligent Agents:

imple Reflex Agent :-

* It acts only on the current inputs.
* Uses predifined rules.
* It doesn't known about previous state.

Working in Smart Home:

-> If motion is detected - TURN ON Lights
-> If no motion - TURN OFF lights
-> If one temperature is above 30°C - TURN ON AC.

Advantages:

-> Fast Response
-> Easy to implement
-> Low Computational Cost.

Disadvantages:

-> May makes incorrect devisions in the complex situations.
-> No memory

2) Model- Agent Based Agent:

* It maintains an internal model of the Environment.
* It also remembers the previous activities.

Working

-> Remembers whether someone is already in the room

→ keeps tracking the Room Occupancy

→ Adjusting the lights with previous sensor data.

Advantages:

→ Better decision making

→ works even with incomplete sensor data.

Disadvantages:

→ Requires memory

→ more complex to Implement

③ Goal- Based Agent

* It selects the action that helps achieve a specific Goal.

working:

→ If everyone leaves the house

→ Lock doors

→ Turn off lights

→ Switch AC to Energy saving mode.

Advantages:

→ Plans actions intelligently

→ flexible in different situations.

Disadvantages:

→ Slower due to Planning.

→ Needs search algorithm.

4) Utility Based - Agent

* It chooses actions that provides minimum utility benefits.

Working:

-> maintains 24°c instead of 18°c to save the Electricity while keeping users comfort zone.

Advantages:

-> optimizes Energy efficiency
-> Learns user perfernces.

Disadvantages:

-> Complex utility Calculations
-> Higher Computational tasks

Best choice:

Hybrid Agent [model Based + Goal - Based Agent + utility Based] is more Effective.

Reason:

- model based remembers the previous state
- Goal based Achieves the Comfort
- utility based optimizes the Energy Efficiency.

Conclusion:

A hybrid intelligent agent provide the best balance between comfort. Energy Efficiency and security.